Homework 5

# ADVANCED MATHEMATICAL PHYSICS

PHYS 721

Name : Begad

ID : 100045069

**5.4.4** The operator $\mathcal{L}$ is Hermitian. Show that $\langle \mathcal{L}^2 \rangle \geq 0$, meaning that for all $\psi$ in the space in which $\mathcal{L}$ is defined, $\langle \psi | \mathcal{L}^2 | \psi \rangle \geq 0$.

Let operator $\mathcal{L}$ be Hermitian. Then,

$$\langle \mathcal{L}^2 \rangle = \langle \psi | \mathcal{L}^2 | \psi \rangle = \langle \mathcal{L}^\dagger \psi | \mathcal{L} \psi \rangle = \langle \mathcal{L}\psi | \mathcal{L}\psi \rangle = \int |\mathcal{L}\psi|^2 \, dx \geq 0$$

def. of adjoint

def of Hermitian

inner product

norm $\geq 0$

**5.5.4** The unitary transformation $U$ that converts an orthonormal basis $\{\varphi_i\}$ into the basis $\{\varphi_i'\}$ and the unitary transformation $V$ that converts the basis $\{\varphi_i'\}$ into the basis $\{\chi_i\}$ have matrix representations

$$\mathsf{U} = \begin{pmatrix} i\sin\theta & \cos\theta & 0 \\ -\cos\theta & i\sin\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathsf{V} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & i\sin\theta \\ 0 & \cos\theta & -i\sin\theta \end{pmatrix}.$$

Given the function $f(x) = 3\varphi_1(x) - \varphi_2(x) - 2\varphi_3(x)$,

(a) By applying $\mathsf{U}$, form the vector representing $f(x)$ in the $\{\varphi_i'\}$ basis and then by applying $\mathsf{V}$ form the vector representing $f(x)$ in the $\{\chi_i\}$ basis. Use this result to write $f(x)$ as a linear combination of the $\chi_i$.

(b) Form the matrix products $\mathsf{UV}$ and $\mathsf{VU}$ and then apply each to the vector representing $f(x)$ in the $\{\varphi_i\}$ basis. Verify that the results of these applications differ and that only one of them gives the result corresponding to part (a).

a) Goal: Represent $f(x)$ in the $\{\varphi_i'\}$ & $\{\chi_i\}$ basis

Solution

· Notation: Let $f_\lambda$ be the representation of $f$ in an arbitrary $\lambda$ basis.

Then,

$$f_{\varphi'} = U f_\varphi = \begin{bmatrix} i\sin\theta & \cos\theta & 0 \\ -\cos\theta & i\sin\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \\ -2 \end{bmatrix} = \begin{bmatrix} 3i\sin\theta - \cos\theta \\ -3\cos\theta - i\sin\theta \\ -2 \end{bmatrix}$$

$$f_\chi = V f_{\varphi_i} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & i\sin\theta \\ 0 & \cos\theta & -i\sin\theta \end{bmatrix} \begin{bmatrix} 3i\sin\theta - \cos\theta \\ -3\cos\theta - i\sin\theta \\ -2 \end{bmatrix} = \begin{bmatrix} 3i\sin\theta - \cos\theta \\ -3\cos^2\theta - i\sin\theta\cos\theta - 2i\sin\theta \\ -3\cos^2\theta - i\sin\theta\cos\theta + 2i\sin\theta \end{bmatrix}$$

Therefore, $f(x) = (-\cos\theta + i\,3\sin\theta)\,\chi_1$

$$+ \left[-3\cos^2\theta - i(\sin\theta\cos\theta + 2\sin\theta)\right]\chi_2$$

$$+ \left[-3\cos^2\theta - i(\sin\theta\cos\theta - 2\sin\theta)\right]\chi_3$$

b) Goal : Form UV & VU and show that only VU leads to $f_x = VUf_\varphi$

Solution

$$UV = \begin{bmatrix} i\sin\theta & \cos\theta & 0 \\ -\cos\theta & i\sin\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & i\sin\theta \\ 0 & \cos\theta & -i\sin\theta \end{bmatrix} = \begin{bmatrix} i\sin\theta & \cos^2\theta & i\sin\theta\cos\theta \\ -\cos\theta & i\sin\theta\cos\theta & -\sin^2\theta \\ 0 & \cos\theta & -i\sin\theta \end{bmatrix}$$

$$VU = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & i\sin\theta \\ 0 & \cos\theta & -i\sin\theta \end{bmatrix} \begin{bmatrix} i\sin\theta & \cos\theta & 0 \\ -\cos\theta & i\sin\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} i\sin\theta & \cos\theta & 0 \\ -\cos^2\theta & i\sin\theta\cos\theta & i\sin\theta \\ -\cos^2\theta & i\sin\theta\cos\theta & -i\sin\theta \end{bmatrix}$$

$$UVf_\varphi = \begin{bmatrix} i\sin\theta & \cos^2\theta & i\sin\theta\cos\theta \\ -\cos\theta & i\sin\theta\cos\theta & -\sin^2\theta \\ 0 & \cos\theta & -i\sin\theta \end{bmatrix} \begin{bmatrix} 3 \\ -1 \\ -2 \end{bmatrix} = \begin{bmatrix} -\cos^2\theta + i(3\sin\theta - 2\sin\theta\cos\theta) \\ -3\cos\theta + 2\sin^2\theta - i\sin\theta\cos\theta \\ -\cos\theta + 2i\sin\theta \end{bmatrix} \neq f_x$$

$$VUf_\varphi = \begin{bmatrix} i\sin\theta & \cos\theta & 0 \\ -\cos^2\theta & i\sin\theta\cos\theta & i\sin\theta \\ -\cos^2\theta & i\sin\theta\cos\theta & -i\sin\theta \end{bmatrix} \begin{bmatrix} 3 \\ -1 \\ -2 \end{bmatrix} = \begin{bmatrix} -\cos\theta + i3\sin\theta \\ -3\cos^2\theta - i(\sin\theta\cos\theta + 2\sin\theta) \\ -3\cos^2\theta - i(\sin\theta\cos\theta - 2\sin\theta) \end{bmatrix} = f_x$$

**5.5.5** Three functions which are orthogonal with unit weight on the range $-1 \leq x \leq 1$ are $P_0 = 1$, $P_1 = x$, and $P_2 = \frac{3}{2}x^2 - \frac{1}{2}$. Another set of functions that are orthogonal and span the same space are $F_0 = x^2$, $F_1 = x$, $F_2 = 5x^2 - 3$. Although much of this exercise can be done by inspection, write down and evaluate all the integrals that lead to the results when they are obtained in terms of scalar products.

(a) Normalize each of the $P_i$ and $F_i$.

(b) Find the unitary matrix $\mathsf{U}$ that transforms from the normalized $P_i$ basis to the normalized $F_i$ basis.

(c) Find the unitary matrix $\mathsf{V}$ that transforms from the normalized $F_i$ basis to the normalized $P_i$ basis.

(d) Show that $\mathsf{U}$ and $\mathsf{V}$ are unitary, and that $\mathsf{V} = \mathsf{U}^{-1}$.

(e) Expand $f(x) = 5x^2 - 3x + 1$ in terms of the **normalized** versions of both bases, and verify that the transformation matrix $\mathsf{U}$ converts the $P$-basis expansion of $f(x)$ into its $F$-basis expansion.

a) Goal & Normalize $P_i$ and $F_i$

Solution

Definition of inner product is given & $\langle f | g \rangle = \int_{-1}^{1} f^* g \, dx$

Normalized $P_i$'s $(P_i')$ &

$$P_1' = \frac{P_1}{\langle P_1 | P_1 \rangle^{1/2}} = \frac{1}{\int_{-1}^{1} 1 \cdot 1 \, dx^{1/2}} = \frac{1}{\sqrt{2}}$$

$$P_2' = \frac{P_2}{\langle P_2 | P_2 \rangle^{1/2}} = \frac{x}{\int_{-1}^{1} x \cdot x \, dx^{1/2}} = \frac{x}{\frac{x^3}{3}\Big|_{-1}^{1}{}^{1/2}} = \sqrt{\frac{3}{2}}\, x$$

$$P_3' = \frac{P_3}{\langle P_3 | P_3 \rangle^{1/2}} = \frac{\frac{3}{2}x^2 - 1/2}{\int_{-1}^{1} \frac{9}{4}x^4 - \frac{3}{2}x^2 + \frac{1}{4} \, dx^{1/2}} = \frac{\frac{3}{2}x^2 - 1/2}{\frac{9}{20}x^5 - \frac{x^3}{2} + \frac{x}{4}\Big|_{-1}^{1}{}^{1/2}} = \sqrt{\frac{5}{2}}\left(\frac{3}{2}x^2 - \frac{1}{2}\right)$$

Normalized $F_i$'s $(F_i')$ &

$$F_1' = \frac{F_1}{\langle F_1 | F_1 \rangle^{1/2}} = \frac{x^2}{\int_{-1}^{1} x^4 \, dx^{1/2}} = \frac{x^2}{\frac{x^5}{5}\Big|_{-1}^{1}{}^{1/2}} = \sqrt{\frac{5}{2}}\, x^2$$

$$F_2' = \frac{F_2}{\langle F_2 | F_2 \rangle^{1/2}} = \frac{x}{\int_{-1}^{1} x^2 \, dx^{1/2}} = \frac{x}{\frac{x^3}{3}\Big|_{-1}^{1}{}^{1/2}} = \sqrt{\frac{3}{2}}\, x$$

$$F_3' = \frac{F_3}{\langle F_3 | F_3 \rangle^{1/2}} = \frac{5x^2 - 3}{\int_{-1}^{1} 25x^4 - 30x^2 + 9 \, dx^{1/2}} = \frac{5x^2 - 3}{5x^5 - 10x^3 + 9x\Big|_{-1}^{1}{}^{1/2}} = \frac{1}{2\sqrt{2}}(5x^2 - 3)$$

b) Goal : Find $U$ such that $c_F = U c_P$ for coefficients of a function $f$ in basis $P$ and $F$.

Solution

We try to find coefficients $u_{ij}$ such that :

$$c_{F_i} = \sum_j u_{ij} c_{P_j} \quad \text{where} \quad c_{F_i} = \langle F_i | \psi \rangle \;,\; c_{P_j} = \langle P_j | \psi \rangle$$

$$\langle F_i | = \sum_j u_{ij} \langle P_j | \quad \Rightarrow \quad u_{ij} = \langle F_i | P_j \rangle \quad \text{to make things consistent}$$

Therefore,

$$U = \begin{bmatrix} \langle F_1 | P_1 \rangle & \langle F_1 | P_2 \rangle & \langle F_1 | P_3 \rangle \\ \langle F_2 | P_1 \rangle & \langle F_2 | P_2 \rangle & \langle F_2 | P_3 \rangle \\ \langle F_3 | P_1 \rangle & \langle F_3 | P_2 \rangle & \langle F_3 | P_3 \rangle \end{bmatrix}$$

$$= \begin{bmatrix} \int_{-1}^{1} \frac{\sqrt{5}}{2} x^2 \, dx & \int_{-1}^{1} \frac{\sqrt{15}}{2} x^3 \, dx & \int_{-1}^{1} \frac{5}{2} \left( \frac{3}{2} x^4 - \frac{x^2}{2} \right) dx \\ \int_{-1}^{1} \frac{\sqrt{3}}{2} x \, dx & \int_{-1}^{1} \frac{3}{2} x^2 \, dx & \int_{-1}^{1} \frac{\sqrt{15}}{2} \left( \frac{3}{2} x^3 - \frac{x}{2} \right) dx \\ \int_{-1}^{1} \frac{1}{4} (5x^2 - 3) \, dx & \int_{-1}^{1} \frac{\sqrt{3}}{4} (5x^3 - 3x) \, dx & \int_{-1}^{1} \frac{\sqrt{5}}{4} \left( \frac{15}{2} x^4 - 7x^2 + \frac{3}{2} \right) dx \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\sqrt{5}}{3} & 0 & 2/3 \\ 0 & 1 & 0 \\ -2/3 & 0 & \sqrt{5}/3 \end{bmatrix}$$

c) Goal : Find $V$ such that $c_P = V c_F$ for coefficients of a function $f$ in basis $P$ and $F$.

Solution

We try to find coefficient $v_{ij}$ such that :

$$c_{P_i} = \sum_j v_{ij} c_{F_j} \quad \Rightarrow \quad v_{ij} = \langle P_i | F_j \rangle = \langle F_j | P_i \rangle = u_{ji}$$

$P, F$ are real.

We can clearly see that $V = U^T$ since $v_{ij} = u_{ji}$ (we expected this since $U$ is unitary $U^{-1} = U^+$)

Therefore,

$$V = \begin{bmatrix} \frac{\sqrt{5}}{3} & 0 & -2/3 \\ 0 & 1 & 0 \\ 2/3 & 0 & \sqrt{5}/3 \end{bmatrix}$$

d) Simply $P = VF = VUP \Rightarrow VU = \underline{I} \Rightarrow V = U^{-1}$

We also got from c) that $V = U^{\dagger}$

Since, $U^{\dagger} = V = U^{-1}$ then $U$ and $V$ are unitary

e) <u>Goal</u>: Write $f(x)$ as $\sum_i^3 c_P P_i(x)$ and $\sum_i^3 c_F F_i(x)$. Then, use $U$ to get $c_P$ from $c_F$

<u>Solution</u>

We can find $c_P$'s and $c_F$'s using scalor products. But, let's proceed by inspection

↖ rebelling against instructions

$P_0 = \frac{1}{\sqrt{2}}$ , $P_1 = \sqrt{\frac{3}{2}}\,x$ , $P_2 = \sqrt{\frac{5}{2}}\left(\frac{3}{2}x^2 - \frac{1}{2}\right)$

The focus here will be on the $x$ ($P_1$) & $x^2$ ($P_2$) term since we only have one of them across the basis.

$$f(x) = 5x^2 - 3x + 1$$

$$= \frac{2\sqrt{2}\cdot\sqrt{5}}{3}P_2 - \sqrt{2}\cdot\sqrt{3}\,P_1 + A\,P_0$$

$$= \frac{2\sqrt{10}}{3}P_2 - \sqrt{6}\,P_1 + \frac{8\sqrt{2}}{3}P_0$$

} $1 = \frac{2\sqrt{10}}{3}\cdot\frac{\sqrt{5}}{\sqrt{2}}(-1/2) + \frac{A}{\sqrt{2}} \Rightarrow A = \sqrt{2}(1 + 5/3) = \frac{8\sqrt{2}}{3}$

$$\Rightarrow C_P = \begin{pmatrix} \frac{8\sqrt{2}}{3} \\ -\sqrt{6} \\ \frac{2\sqrt{10}}{3} \end{pmatrix}$$

$F_0 = \sqrt{\frac{5}{2}}\,x^2$ , $F_1 = \sqrt{\frac{3}{2}}\,x$ , $F_2 = \frac{1}{2\sqrt{2}}(5x^2 - 3)$

The focus here will be on the constant ($F_2$) & $x$ ($F_1$) term since we only have one of them across the basis.

$$f(x) = 5x^2 - 3x + 1$$

$$= -\frac{2\sqrt{2}}{3}F_2 - \sqrt{6}\,F_1 + A\,F_0$$

$$= -\frac{2\sqrt{2}}{3}F_2 - \sqrt{6}\,F_1 + \frac{4\sqrt{10}}{3}F_0$$

} $A\sqrt{\frac{5}{2}} - \frac{2\sqrt{2}}{3}\cdot\frac{1}{2\sqrt{2}}\cdot 5 = 5 \Rightarrow A = \frac{\sqrt{2}}{\sqrt{5}}(5 + 5/3) = \frac{4\sqrt{10}}{3}$

$$\Rightarrow C_F = \begin{pmatrix} \frac{4\sqrt{10}}{3} \\ -\sqrt{6} \\ -\frac{2\sqrt{2}}{3} \end{pmatrix}$$

We recall that if we want to transform the coordinates from basis P to F. We use transformation U. $c_F = U c_P$

Let's confirm it!

$$U_{CP} = \begin{bmatrix} \frac{\sqrt{5}}{3} & 0 & 2/3 \\ 0 & 1 & 0 \\ -2/3 & 0 & \sqrt{5}/3 \end{bmatrix} \begin{bmatrix} 8\sqrt{2}/3 \\ -\sqrt{6} \\ 2\sqrt{10}/3 \end{bmatrix} = \begin{bmatrix} 8\sqrt{10}/9 + 4\sqrt{10}/9 \\ -\sqrt{6} \\ -\frac{16\sqrt{2}}{9} + \frac{10\sqrt{2}}{9} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{4\sqrt{10}}{3} \\ -\sqrt{6} \\ -\frac{2\sqrt{2}}{3} \end{bmatrix} = C_F$$

**5.7.2** The Pauli matrices

$$\boldsymbol{\sigma}_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \boldsymbol{\sigma}_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \boldsymbol{\sigma}_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

have commutator $[\boldsymbol{\sigma}_1, \boldsymbol{\sigma}_2] = 2i\boldsymbol{\sigma}_3$. Show that this relationship continues to be valid if these matrices are transformed by

$$\mathsf{U} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}.$$

Goal : Commutator Relation $[\sigma_1, \sigma_2] = 2i\sigma_3$ is invariant under transformation U.

Solution

We suspect that U is unitary. (It is since it's the 2D-Rotation Matrix)

Let's prove it :

transpose

$$UU^{\dagger} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I \quad \Rightarrow \text{U is unitary}$$

Let A be an operator / matrix transformed under a unitary matrix U.

Then, the transformed matrix $A' = UAU^{-1}$

$$\sigma_1' = U\sigma_1 U^{-1} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} \sin\theta & \cos\theta \\ \cos\theta & -\sin\theta \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

$$\sigma_2' = U\sigma_2 U^{-1} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} i\sin\theta & -i\cos\theta \\ i\cos\theta & i\sin\theta \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

$$\sigma_3' = U\sigma_3 U^{-1} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ -\sin\theta & -\cos\theta \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

$$\sigma_1' = \begin{bmatrix} \sin 2\theta & \cos 2\theta \\ \cos 2\theta & -\sin 2\theta \end{bmatrix} \qquad \sigma_2' = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \qquad \sigma_3' = \begin{bmatrix} \cos 2\theta & -\sin 2\theta \\ -\sin 2\theta & -\cos 2\theta \end{bmatrix}$$

$$[\sigma_1', \sigma_2'] = \sigma_1'\sigma_2' - \sigma_2'\sigma_1'$$

$$= \begin{bmatrix} \sin 2\theta & \cos 2\theta \\ \cos 2\theta & -\sin 2\theta \end{bmatrix}\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} - \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}\begin{bmatrix} \sin 2\theta & \cos 2\theta \\ \cos 2\theta & -\sin 2\theta \end{bmatrix}$$

$$= \begin{bmatrix} i\cos 2\theta & -i\sin 2\theta \\ -i\sin 2\theta & -i\cos 2\theta \end{bmatrix} - \begin{bmatrix} -i\cos 2\theta & i\sin 2\theta \\ i\sin 2\theta & i\cos 2\theta \end{bmatrix}$$

$$= 2i \begin{bmatrix} \cos 2\theta & -\sin 2\theta \\ -\sin 2\theta & -\cos 2\theta \end{bmatrix} = 2i\sigma_3' \qquad \text{as required} \quad \blacksquare$$

* We could have used the properties of unitary matrices to prove this too.

$$[\sigma_1', \sigma_2'] = \sigma_1'\sigma_2' - \sigma_2'\sigma_1'$$

$$= U\sigma_1 U^{-1} U\sigma_2 U^{-1} - U\sigma_2 U^{-1} U\sigma_1 U^{-1}$$

$$= U\sigma_1\sigma_2 U^{-1} - U\sigma_2\sigma_1 U^{-1}$$

$$= U(\sigma_1\sigma_2 - \sigma_2\sigma_1)U^{-1}$$

$$= U(2i\sigma_3)U^{-1}$$

$$= 2i\, U\sigma_3 U^{-1} = 2i\sigma_3'$$

**6.2.15** Describe the geometric properties of the surface

$$x^2 + 2xy + 2y^2 + 2yz + z^2 = 1.$$

How is it oriented in 3-D space? Is it a conic section? If so, which kind?

Goal : Uncover the geometric properties of $f(x,y,z)$

Solution

We suspect that $f(x,y,z)$ is some conic section since it is quadratic in $x$, $y$ and $z$. However, it is not in a form that is clear for us to tell.

Paraboloid  Sphere  ellipsoid  cylinder  elliptic paraboloid  ?

Maybe studying the eigenvalues & vectors of the quadratic form could tell us more.

Let $\vec{x}^T A \vec{x}$ be the quadratic matrix form of $f(x,y,z)$

Then,

$$\begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = x^2 + 2xy + 2y^2 + 2yz + z^2$$

we use this $A$ since it is symmetric.

Minigoal : Find $\lambda_i$ for $A$

$$A\vec{x} = \lambda\vec{x} \Rightarrow (A - \lambda I)\vec{x} = 0$$

$$\det \begin{bmatrix} 1-\lambda & 1 & 0 \\ 1 & 2-\lambda & 1 \\ 0 & 1 & 1-\lambda \end{bmatrix} = 0$$

$$(1-\lambda)((2-\lambda)(1-\lambda) - 1) - 1 - \lambda = 0$$

$$(2-\lambda)(1-\lambda)^2 - 2(1-\lambda) = 0$$

$$(1-\lambda)[(2-\lambda)(1-\lambda) - 2] = 0$$

$$(1-\lambda)(\lambda^2 - 3\lambda) = 0$$

$$\lambda(1-\lambda)(\lambda-3) = 0 \Rightarrow \lambda_1 = 0,\ \lambda_2 = 1,\ \lambda_3 = 3$$

Minigoal 8 : Find eigenvectors $\vec{x}_i$ for A

For $\lambda_1 = 0$, $A\vec{x} = \vec{0}$

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x+y \\ x+2y+z \\ y+z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Let $y$ be the auxilliary variable, s.t. $y = a$. Then, 1) $x = -y = -a$ 2) $-a + 2a - a = 0$ ✓

3) $z = -y = -a$

Therefore, $\vec{x}_1$ is of the form $a(1, -1, 1)$.

For $\lambda_1 = 1$, $(A - I)\vec{x} = \vec{0}$

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} y \\ x+y+z \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Let $x$ be the auxilliary variable, s.t. $x = a$. Then, 1) $y = 0$ 2) $a + 0 + z = 0 \Rightarrow z = -a$.

3) $y = 0$

Therefore, $\vec{x}_2$ is of the form $a(1, 0, -1)$.

For $\lambda_1 = 3$, $(A - 3I)\vec{x} = \vec{0}$

$$\begin{bmatrix} -2 & 1 & 0 \\ 1 & -1 & 1 \\ 0 & 1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2x+y \\ x-y+z \\ y-2z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Let $y$ be the auxilliary variable, s.t. $y = a$. Then, 1) $y = 2x \Rightarrow x = a/2$ 2) $a/2 - a + a/2 = 0$ ✓

3) $y = 2z \Rightarrow z = a/2$

Therefore, $\vec{x}_3$ is of the form $a(1, 2, 1)$.

Awesome! We are in tip top shape.

We recall that our matrix is Symmetric and can be decomposed by a

$$A = Q \Lambda Q^T$$

Such that

orthogonal matrix

$$\begin{bmatrix} | & | & | \\ x_1 & x_2 & x_3 \\ | & | & | \end{bmatrix}$$

↓ eigen vectors are columns

diagonal matrix

$$\begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$$

↑ eigenvalues

This is swell since we can now represent A as a diagonal matrix in the new basis making the transformation simpler

Let $\vec{x} = Q^T \vec{v}$ be our new transformed variables.

Then, $\vec{x}^T A \vec{x} = \vec{v}^T Q A Q^T \vec{v} = \vec{v}^T D \vec{v}$

↑ A is diagonal in the v basis!

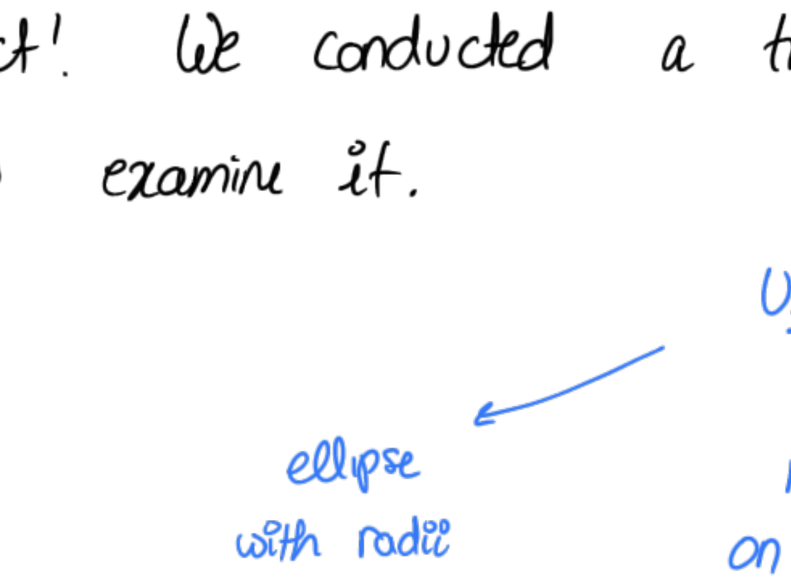
$$\vec{v}^T D \vec{v} = \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = v_2^2 + 3v_3^2 = 1$$

Perfect! We conducted a transformation that took $f(x,y,z) \rightarrow f(v_1, v_2, v_3)$

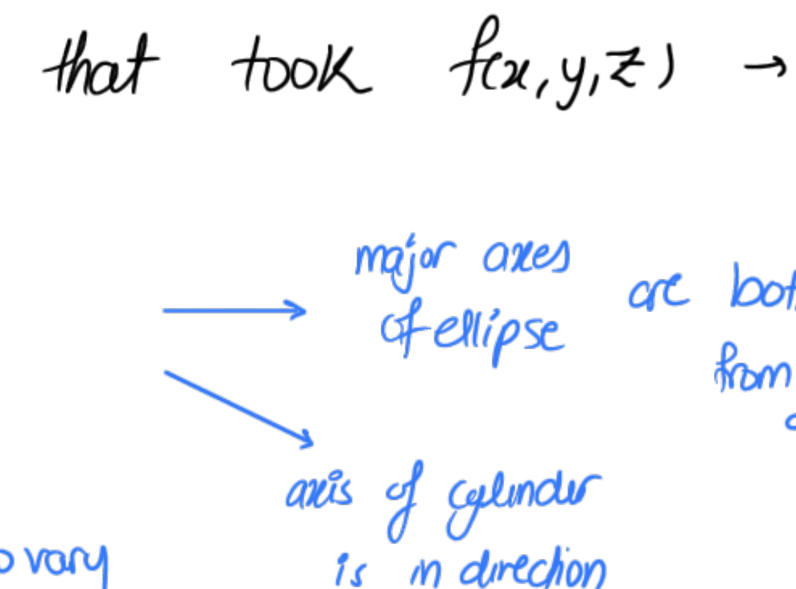
Let's examine it.

$v_2^2 + 3v_3^2 = 1$

- ellipse with radii 1 and $1/\sqrt{3}$
- no dependence on $v_1$ ∴ $v_1$ allowed to vary
  The ellipse is the cross-section of the elliptic cylinder
- major axes of ellipse are both $\vec{x}_2$ & $\vec{x}_3$ from the dependence on $v_2$ and $v_3$
- axis of cylinder is in direction of $\vec{x}_1$ ($\vec{x}_1 \perp$ ellipse crossection)




$f(x,y,z)$




cross-section includes $(0,0,0)$

**6.4.1** Show that the eigenvalues of a matrix are unaltered if the matrix is transformed by a similarity transformation—a transformation that need not be unitary, but of the form given in Eq. (5.78).

This property is not limited to symmetric or Hermitian matrices. It holds for any matrix satisfying an eigenvalue equation of the type $\mathsf{A}\mathbf{x} = \lambda\mathbf{x}$. If our matrix can be brought into diagonal form by a similarity transformation, then two immediate consequences are that:

1. The trace (sum of eigenvalues) is invariant under a similarity transformation.
2. The determinant (product of eigenvalues) is invariant under a similarity transformation.

*Note.* The invariance of the trace and determinant are often demonstrated by using the Cayley-Hamilton theorem, which states that a matrix satisfies its own characteristic (secular) equation.

A matrix $A$ is said to be similar to $A'$ if

$$A' = GAG^{-1} \quad \text{for an invertible matrix } G.$$

From (5.78), we know that similarity transformation follow that

$$A\vec{c} = \vec{b}$$

$$\underbrace{GAG^{-1}}_{A_G \text{ in new } G \text{ basis}} \cdot \underbrace{G\vec{c}}_{c_G \text{ in new } G \text{ basis}} = \underbrace{G\vec{b}}_{b_G \text{ in new } G \text{ basis}}$$

Now, if we were to take $\vec{c} = \vec{x}$ eigenvector

$\vec{b} = \lambda\vec{x}$ eigenvalue × eigenvector then

$$\underbrace{GAG^{-1}}_{A_G} \cdot \underbrace{G\vec{x}}_{\vec{x}_G} = G\lambda\vec{x} = \lambda G\vec{x}$$

$$= \lambda\,\vec{x}_G \quad \rightarrow \lambda \text{ is still the eigenvalue of matrix } A_G \text{ with } \vec{x}_G \text{ eigenvector.}$$